

5.4.2 Identifying Security Threats

خطرے کی جلد پہچان نقصان سے پہلے اقدام ممکن بناتی ہے۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 12 of 17

5.4.2 Identifying Security Threats

LEARN IT

Identifying security threats means **noticing signs** that data or systems may be at risk. Early identification helps you act before serious damage occurs.

Sign / indicator	What it likely means
Slow performance, missing files, or random pop-ups.	Malware or background unauthorised activity.
Failed login attempts or strange login locations.	Attempted unauthorised access .
Emails urging quick action, asking for an OTP, or containing bad grammar / strange links.	Classic phishing .
Unexpected deletion, copying or movement of records.	Data theft or insider misuse.
Sudden massive spike in server requests or a very slow website.	An ongoing DoS attack.

The first line of defence is also simple behaviour: avoid **unsafe downloads, cracked software, unverified websites and infected USB drives**.

WORKED EXAMPLE

Worked example: read the signs in a college computer lab in Khairpur.

1. Three PCs show pop-ups and Word files have vanished after students used a "game" USB. → Treat as **malware**. Disconnect from the network; run antivirus; stop USB use.
2. Google Classroom emails: "Your account will be deleted in 30 minutes. Confirm password here." Link spelling is goOgle. → **Phishing**. Do not click; report to the teacher.
3. The e-portal that was fine at 8:00 am is unreachable at 8:05; the technician sees a sudden spike in requests. → Suspect **DoS**.
4. Nadia's Gmail security page shows a login from another country at 4 a.m. → **Unauthorised access** attempt. Change password; turn on MFA.

REMEMBER IT

Slow/pop-ups → **malware**. **Odd logins** → **unauthorised access**. **Urgent OTP email** → **phishing**. **Files moving** → **theft**. **Traffic spike** → **DoS**.

COMMON MISTAKE

Mistake: Ignoring early signs ("the PC is just old") until the ransomware note appears.

Fix: One unusual login or one OTP email is enough to investigate. Early action is cheaper than recovery.

خطرے کی جلد پہچان نقصان سے پہلے اقدام ممکن بناتی ہے۔ سست روی، غائب فائلیں یا پاپ اپ میلویئر کی علامت ہیں؛ عجیب لاگ ان غیر مجاز رسائی کی؛ فوری او ٹی پی والی ای میل فہشنگ کی؛ ریکارڈ کا اچانک منتقل ہونا ڈیٹا چوری کی؛ سرور پر درخواستوں کا دھماکہ ڈو ایس کی۔ کریک سافٹ ویئر، نامعلوم یو ایس بی اور غیر تصدیق شدہ ویب سائٹوں سے بچیں۔

CHECK YOURSELF

1. Random pop-ups and missing files most likely indicate what?
2. An email with bad grammar that demands your OTP is a sign of what?
3. Name four unsafe habits that invite these threats.

Answers: 1. Malware (or background unauthorised activity). 2. Phishing. 3. Unsafe downloads, cracked software, unverified websites, infected USB drives.

Next lecture: 5.4.3 Apply Mitigation Techniques